AI.306

Biologically Inspired Robotics

COURSE OVERVIEW

Robots modeled on nature: locomotion, sensing, and control principles from animals and plants.

1

SESSION

60 MINUTES

ORIENTATION: LEARNING FROM LIFE

KEY TOPICS

- Bio-inspiration vs. biomimetics
- Why nature is a good teacher
- Organisms as models
- Course scope

TEACHING FOCUS

Mine biology for robot design ideas.

KEY TAKEAWAY Evolution has already solved mobility.

2

SESSION

⏱ 60 MINUTES

BIO-INSPIRED LOCOMOTION

KEY TOPICS

- Snake, fish, and insect gaits
- Muscle-like actuators
- Compliant mechanisms
- Soft robotics

TEACHING FOCUS

Move like living organisms.

▲ **KEY TAKEAWAY** Nature's movers are efficient and adaptable.

3

SESSION

 60 MINUTES

BIO-INSPIRED SENSING

KEY TOPICS

- Vision systems: insect eyes
- Whiskers and tactile
- Echolocation and lateral line
- Chemical sensing

TEACHING FOCUS

Sense like animals do.

▲ **KEY TAKEAWAY** Biological sensors are tuned and efficient.

4

SESSION

60 MINUTES

NEUROMORPHIC CONTROL

KEY TOPICS

- Spiking neural networks
- Central pattern generators
- Reflexes and reflexes
- Bio-control architectures

TEACHING FOCUS

Control like a nervous system.

▲ **KEY TAKEAWAY** Brains control bodies efficiently.

5

SESSION

60 MINUTES

SWARM & SOCIAL BIOLOGY

KEY TOPICS

- Ant colonies and bee swarms
- Flocking and schooling
- Territoriality
- Collective behavior

TEACHING FOCUS

Coordinate like social animals.

KEY TAKEAWAY Biology excels at decentralized coordination.

BIOLOGICALLY INSPIRED ROBOTICS

6

SESSION

60 MINUTES

BIO-ROBOTICS & REVIEW

KEY TOPICS

- Bio-robot prototype
- Course recap and assessment
- Toward machine vision (AI.307)
- Design review

TEACHING FOCUS

Build robots inspired by life.

KEY TAKEAWAY Nature is the ultimate robotics lab.